Convergence Tests

ANSWER KEY

The ratio and comparison tests

1 Use the ratio test to determine whether $\sum_{n=1}^{\infty} \frac{n}{2^n}$ converges.
$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\frac{n+1}{2n}=\frac{1}{2}<1$: converges.

2 Use the ratio test on $\sum_{n=1}^{\infty} \frac{n!}{n^n}$.
$\frac{a_{n+1}}{a_n} = \frac{n^n}{(n+1)^n} = \left(\frac{n}{n+1}\right)^n \to \frac{1}{e} < 1$: converges.

3 Use the comparison test to determine whether $\sum_{n=1}^{\infty} \frac{1}{n^2+3}$ converges, by comparing to $\sum \frac{1}{n^2}$.
$0 < \frac{1}{n^2+3} < \frac{1}{n^2}$ for all $n \geq 1$, and $\sum \frac{1}{n^2}$ is a convergent p-series ($p = 2 > 1$), so by comparison the given series converges.

Alternating series and the p-series test

4 Determine whether each p-series converges or diverges:

a $\sum_{n=1}^{\infty} \frac{1}{n^3}$ Converges ($p = 3 > 1$).

b $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$ Diverges ($p = \frac{1}{2} \leq 1$).

5 Determine whether $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$ converges, and classify it as absolutely convergent, conditionally convergent, or divergent.
The terms $\frac{1}{n}$ decrease to 0, so by the alternating series test the series converges. But $\sum \frac{1}{n}$ (absolute values) diverges — so the series is conditionally convergent.

The integral test

6 Use the integral test to determine whether $\sum_{n=1}^{\infty} \frac{1}{n \ln n}$ (starting at $n = 2$) converges or diverges, using $\int_2^{\infty} \frac{1}{x \ln x}\,dx$.
Let $u = \ln x$: $\int \frac{1}{x \ln x} dx = \ln|\ln x| + C$. $\int_2^{\infty} \frac{dx}{x \ln x} = \lim_{b\to\infty}[\ln(\ln x)]_2^b = \infty$: the integral diverges, so by the integral test the series diverges.

7 Use the integral test to confirm that $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges, using $\int_1^{\infty} \frac{1}{x^2}\,dx$.
$\int_1^{\infty} \frac{1}{x^2} dx = \lim_{b\to\infty}\left[-\frac{1}{x}\right]_1^b = 1$, a finite value, so by the integral test the series converges.

Choosing the right test

8 For each series, name the most efficient test to apply (ratio, comparison, p-series, or alternating series), without fully evaluating:

a $\sum\frac{2^n}{n!}$ Ratio test (factorial in denominator is a strong signal).

b $\sum\frac{(-1)^n}{\sqrt{n}}$ Alternating series test (terms decrease to 0).

c $\sum\frac{1}{n^4}$ p-series test directly ($p = 4 > 1$: converges).

The limit comparison test

9 Use the limit comparison test to determine whether $\sum_{n=1}^{\infty}\frac{n+1}{n^3+2}$ converges, comparing to $\sum\frac{1}{n^2}$.
$\lim_{n\to\infty}\frac{(n+1)/(n^3+2)}{1/n^2} = \lim_{n\to\infty}\frac{n^3+n^2}{n^3+2} = 1$, a finite positive value. Since $\sum\frac{1}{n^2}$ converges ($p = 2 > 1$), by the limit comparison test the given series also converges.

10 Use the limit comparison test on $\sum^{\infty}\frac{3n^2-1}{n^4+2n}$, comparing to $\sum\frac{1}{n^2}$.
$\lim_{n\to\infty}\frac{(3n^2-1)/(n^4+2n)}{1/n^2} = \lim_{n\to\infty}\frac{3n^4-n^2}{n^4+2n} = 3$, finite and positive. Since $\sum\frac{1}{n^2}$ converges, so does the given series.